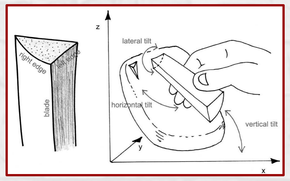
Predicting Object Provenance based on the Appearance of Cuneiform Signs

Bryce Hoenigman
Research Advisor: Aren Wilson-Wright

RESEARCH
Computing Center

What is Cuneiform?

- **What**: Considered the oldest writing system, used for many ancient languages such as Akkadian, Sumerian, and Hittite
- **How**: Imprinted by indenting styli onto wet clay surfaces resulting in a 3D effect



- **When**:~3000 BCE - 300 CE
- **Where:** Mesopotamia

Key: the cuneiform script evolved over the time and regions it was used in → that variation is what this project aims to study

Overview

- **Goal:** predict where an unprovenanced cuneiform-inscribed object was written from the shape of its signs (hypothesis generation for philologists)
- **Motivation:**
  - Poor archaeological practices → provenance is often unclear
  - Situating texts within cultural contexts informs interpretation
  - Learning about scribal conventions

Dataset

- **eBL Signs Dataset** 158,946 single-sign images from 9,276 unique tablets
- **Findspots rebuilt** from CDLI + eBL Fragmentarium: 80.4% of images labeled



Methods

1. Encoding Sign Images into Dense Feature Representations

- **Preprocessing:** standardized scale, color, and normalized to ImageNet
- **Problem**: generic encoders (ResNet-50, DINOv2) sorted signs by tablet material (clay vs stone) not shape of signs, even after grayscale + equalization

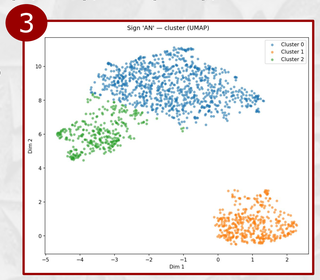


- **Solution**: Fine-tuned CNN encoders (ResNet101 and EfficientNetV2-S with TrivialAugmentWide) on 266-class sign identification, using about 140k images of signs with at least 50 attestations → yielded 84.3% accuracy and 0.764 macro-F1 → landed on EfficientNetV2-S, 1,280-dimensional embedding per sign

2. Exploring Variation: Dimensionality Reduction and Clustering

- **Old Babylonian** (ca. 1900–1600 BCE): 34,886 images, 413 tablets.
- **Six high-frequency signs**: AN (n = 1,689), A (1,507), NA (819), MA (694), DIŠ (658), and GAR (173).
- Embeddings standardized and projected **3 ways**: (1) PCA (2) t-SNE (3) UMAP

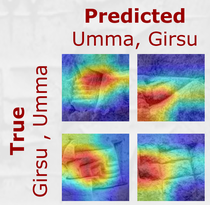
3



- **Model**: EfficientNetV2-S, TrivialAugmentWide, early stopping, weighted cross-entropy, and Optuna hyperparameter search → 84.3% validation accuracy, 0.764 macroF1
- **Clustering:**
  - Each point = one sign occurrence (1,280-d embedding) → each sign projected to 2D and clustered separately to find variant forms (e.g. AN clustered into monumental, cursive, and conjunct forms)
  - Colored by k-means for each projection → compared agreement and whether groups correspond to interpretable paleographic categories

3. Generating Provenance Predictions (full corpus)

- 70/15/15 splits at the tablet level so no single text leaks across sets
- **Model**: EfficientNet-B0 with transfer learning, early stopping + optuna search, macro-F1 as primary metric
- **Handling class imbalance**: inverse-frequency weighted cross-entropy
- **Four tasks** (progressively finer): (**1**) north/south within the Neo-Babylonian period, (**2**) north/south/periphery, (**3**) 16 cities, (**4**) 5 Ur III sites
- Held **period constant** (Neo-Babylonian and Ur-III - both well represented) to show the model learns region, not era
- **Interpretability**: Grad-CAM (final convolutional layer) → attention diffuseness (% pixels over 50% peak) and whether confused cities share attention patterns (within- and cross-class IoU and cosine similarity)



Results

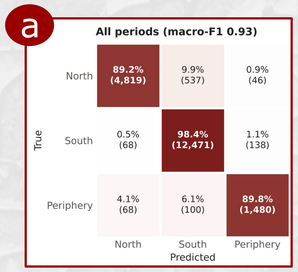
Classification:

1. **N/S/Periphery**: mF1=0.93, acc=95% (a)
2. **N/S-NB only**: mF1=0.95, acc=98% (b)
3. **16 cities**: mF1=0.68 (baseline=0.017) (c)

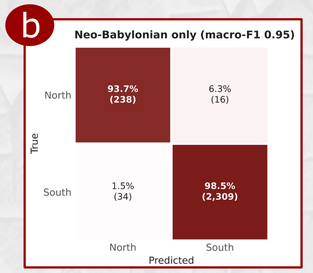
   Strongest: Assur(0.99), Kalhu(0.95), Babylon(0.92)

   Weakest: Girsu(0.36), Ur(0.21), Sippar-Yahurum(0.09), Nippur absorbs most errors
4. **5 Ur-III sites**: mF1=0.68, Girsu and Umma confused, but did well without Nippur (mF1=0.892)

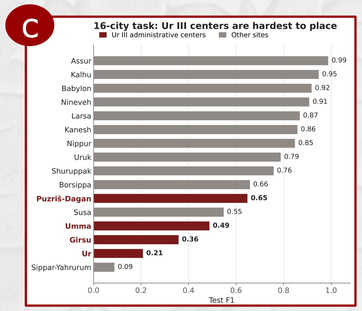
a



b



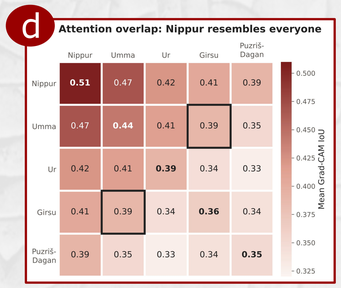
c



Grad-CAM:

- Correct predictions had tighter attention than incorrect ones (0.31 vs 0.41 spread)
- Nippur had highest in-class consistency (IoU 0.51), Girsu had the lowest (IoU 0.36)
- Girsu/Umma attention similarity was 6/10 pairs, the pairs involving Nippur were the most similar (d)

d



Discussion

- General findspot predictable from sign-level visual features alone (survives period controls ⇒ differences in scribal tradition)
- Accuracy falls with standardization (like Ur III administrative centers)

Limitations:

- Limited data, especially with smaller classes
- Grad-CAM is of limited use on pre-cropped sign images since the sign always fills the center

Future direction:

- Aggregate sign predictions into a more informed tablet-level prediction

Acknowledgements

- **Dr. Aren Wilson-Wright** for guidance and mentorship throughout this project.
- Computing was supported by the **University of Chicago Research Computing Center (RCC)**, whose Midway3 cluster provided the GPU resources used for model training.
- **Image credits:** Cuneiform stylus illustration from "The Cuneiform Stylus" (Academia.edu); sign/tablet images from the *Journal of Open Humanities Data* article (doi:10.5334/johd.503);